Indian Journal of Cryptography and Network Security (IJCNS)
ISSN: 2582-9238 (Online), Volume-1 Issue-3, May 2022

Reliability of Trust Management Systems in Cloud Computing

Pooja Goyal, Sukhvinder Singh Deora

Abstract*: Cloud computing is an innovation that conveys administrations like programming, stage, and framework over the web. This computing structure is wide spread and dynamic, which chips away at the compensation per-utilize model and supports virtualization. Distributed computing is expanding quickly among purchasers and has many organizations that offer types of assistance through the web. It gives an adaptable and on-request administration yet at the same time has different security dangers. Its dynamic nature makes it tweaked according to client and supplier's necessities, subsequently making it an outstanding benefit of distributed computing. However, then again, this additionally makes trust issues and or issues like security, protection, personality, and legitimacy. In this way, the huge test in the cloud climate is selecting a perfect organization. For this, the trust component assumes a critical part, in view of the assessment of QoS and Feedback rating. Nonetheless, different difficulties are as yet present in the trust the board framework for observing and assessing the QoS. This paper talks about the current obstructions present in the trust framework. The objective of this paper is to audit the available trust models. The issues like insufficient trust between the supplier and client have made issues in information sharing likewise tended to here. Besides, it lays the limits and their enhancements to help specialists who mean to investigate this point.

Keywords*: Malicious Users, Subjectivity, Cloud Environment, Service Level Agreement, Reputation System, Quality of Service (QoS), Credibility.

I. INTRODUCTION

In the present situation, Cloud Computing is the cerebrum of web figuring. Disseminated figuring has acquired titanic changes the administrations given by the web. Distributed computing gives us a great deal of advantages like asset pooling, on-request benefits, expansive organization access, administration estimation, and fast flexibly. Distributed computing is fundamentally characterized as an information stockpiling, bunch, or supercomputer that conveys figuring assets through the web, on-request from a far off area as opposed to harping on one's workspace, cell phones, or PC. Distributed computing can be carried out through various sending models like public, private, local area, and half breed[1]. The functioning guideline of distributed computing is to make handling and capacity framework free of appropriate setting for the cloud client and offers dependable and effective types of assistance. Distributed computing gives us many administrations like framework, stage, programming, stockpiling, master, climate, information, and security on-request at a lower cost. Be that as it may, each beneficial thing accompanies a few entanglements. The blemish of distributed computing remains with its current circumstance whose nature is dynamic, perplexing, non-straightforward, and open access. Customers have a dubious outlook on what emerge to their information once it goes into the cloud. They ponder who can get to their information and how it will be put away, record, shared, and utilized. Also, they perceive like they lose command of their data[2]. Because of this, shoppers feel shaky at the hour of utilizing cloud administrations. In addition, the determination of any cloud administration relies upon the QoS and capacity performed by specialist co-ops. Yet, the significant issue is that we can't quantify QoS precisely as the idea of the cloud is dynamic, and furthermore the prerequisite of buyers changes according to their requirements. Likewise, the greatest wellspring of estimating the presentation of cloud administration is criticism given by its authentic clients. Yet, this input is influenced by a vindictive element. To deal with this whole hindrance, the idea of trust is utilized between a purchaser and a supplier. For building this confidence, a Trust the executives framework is utilized. Distributed computing gives a pool of figuring assets (organization, working framework, application, stockpiling, server) of top caliber with lower cost over the cloud through the web [1]. It chips away at the compensation according to utilize idea that can be gotten to from any spot, through any gadget whatsoever time SLA containing all the data identified with the concurrence with characterized QoS (Quality of administration) among shoppers and suppliers [2]. Distributed computing is known as the aftereffect of an advancement of the boundless reception of virtualization, administration arranged engineering, autonomic, and utility processing[3]. The Details of the area of framework or part gadgets are obscure to a large portion of the end-clients, as the client doesn't have to completely comprehend or control the innovation foundation that upholds their registering activities[4][5][6] Trust is a shared comprehension between two substances who need to speak with one another for business purposes. The foundation of any relationship is trust. The dependability of the trust is partitioned into abstract and objective [8].

Manuscript received on November 02, 2021.
Revised Manuscript received on November 20, 2021.
Manuscript published on May 10, 2021.
* Correspondence Author
Ms. Pooja Goyal*, Assistant Professor, Department of Computer Science & Application, Maharshi Dayanand University, Rohtak (Haryana) India. E-mail: poojagoyal895@gmail.com
Dr. Sukhvinder Singh Deora, Assistant Professor, Department of Computer Science & Application, Maharshi Dayanand University, Rohtak (Haryana) India.

© The Authors. Published by Lattice Science Publication (LSP). This is an open access article under the CC BY-NC-ND license (http://creativecommons.org/licenses/by-nc-nd/4.0/)

Target trust is an examination among dispensed and assessed administrations through assistance level understanding. Subsequently, the trust of the specialist organization increments when it offers types of assistance as per the arrangement. Abstract trust is identified with criticism appraisals presented by different help buyers. It relies upon buyer's information and inclination about administrations during their collaboration [9]. On the off chance that a customer grants a specialist organization with great input, the trust esteem increments. In any case, trust is emotional in nature and relies upon the insight and setting of buyers. Yet, some unsettled difficulties are available in this instrument [10].Objective trust is a correlation among apportioned and assessed administrations through assistance level arrangement. Subsequently, the trust of the specialist organization increments when it offers types of assistance as indicated by the arrangement. Abstract trust is identified with criticism appraisals presented by different help purchasers. It relies upon customer's information and inclination about administrations during their connection [9]. In the event that a buyer grants a specialist co-op with great input, the trust esteem increments. Be that as it may, trust is abstract in nature and relies upon the insight and setting of buyers. Yet at the same time, some unsettled difficulties are available in this systemis

The main intent of this article is to

- Brief presentation of distributed computing with its component, sorts of model.
- Draw thoughtfulness regarding numerous significant issues identified with security just as protection in a cloud climate.
- Trust Management framework with its sorts
- Challenges identified with trust assessment instrument and notoriety framework.
- Summary of different trust assessment draws near.

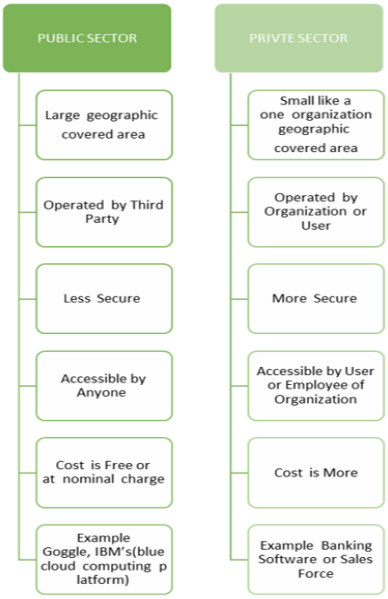
II. CLOUD COMPUTING

The idea of distributed computing appeared in 1950 with the achievement of centralized server PCs, available by means of flimsy/static customers. From that point forward, distributed computing has been developed from stale customers to dynamic ones from programming to administrations [26]. Distributed computing gives us a layout that contains predefined design insights regarding various specialist organizations. It chips away at, pay-more only as costs arise valuing idea. Distributed computing yields numerous monetary and specialized benefits. The primary target of distributed computing is to give fast admittance to adaptable and proficient assets at a lower cost [11]. Besides, it delivers extra virtual space for associations for their authoritative activity or conveys their application. We can likewise characterize distributed computing as a blend of framework registering and administration situated figuring for effective use of assets and giving foundation, programming, and stage as assets to the cloud clients. The two essential pieces of distributed computing are the front end and the back end. The front end is the customer, the cloud administration buyer who has an application and interface (web program) needed for getting to distributed computing stages. The back end is essentially the cloud. It involves data stockpiling, security parts, servers, and virtual machines [12].

A. Cloud Building Deployment Model

Disseminated figuring is otherwise called a two-overlap edged weapon, gives similar administrations and control of the assets to the two customers and aggressors to use for their benefit[7]. Thus, when aggressors get similar regulatory force as cloud clients, they can play out various unfortunate exercises on the client information and trick numerous blameless clients. They can perform various types of assaults, produce treats, appropriate pilfered programming, change buyer's s individual data, or utilize client's information anyplace or discover approaches to moves bugs to cloud customers 2. The idea of distributed computing appeared in 1950 with the achievement of centralized server PCs, available by means of flimsy/static customers. From that point forward, distributed computing has been developed from stale customers to dynamic ones from programming to administrations [26]. Distributed computing gives us a layout that contains predefined design insights regarding various specialist organizations. It chips away at, pay-more only as costs arise valuing idea. Distributed computing yields numerous monetary and specialized benefits. The primary target of distributed computing is to give fast admittance to adaptable and proficient assets at a lower cost [11]. Besides, it delivers extra virtual space for associations for their authoritative activity or conveys their application. We can likewise characterize distributed computing as a blend of framework registering and administration situated figuring for effective use of assets and giving foundation, programming, and stage as assets to the cloud clients. The two essential pieces of distributed computing are the front end and the back end. The front end is the customer, the cloud administration buyer who has an application and interface (web program) needed for getting to distributed computing stages. The back end is essentially the cloud. It involves data stockpiling, security parts, servers, and virtual machines [12].

B. Cloud Building Deployment Model



© Copyright: All rights reserved.

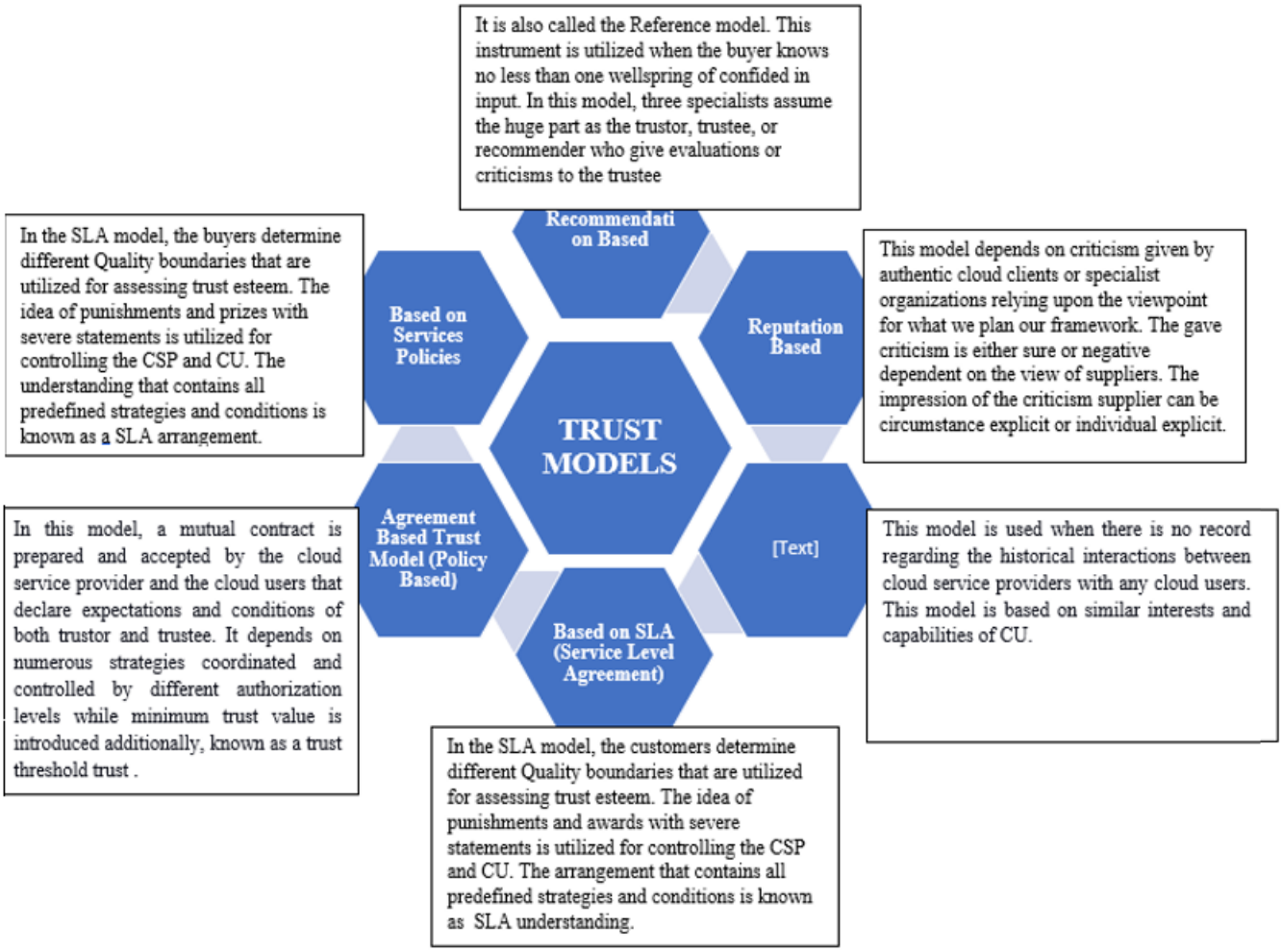
C. Characteristics of cloud computing

- The working procedure of conveyed registering is on-request. Because of this, the client can associate with their administrations according to necessity and control their administrations.
- Cloud clients can get to cloud administrations through any favored medium like PCs, work areas, and cell phones.
- Among the most momentous provisions of distributed computing is quick versatility. By this, assets are given rapidly according to require.
- It chips away at dispersed engineering and offers its types of assistance to the private and public areas through virtualization procedures.
- It chips away at the compensation as-utilize model. It consequently investigates the prerequisite of assets and supports an estimating capacity at some degree of administration. This use of figuring assets is controlled, estimated, and answered to cloud clients and specialist organizations.
- It deals with administration situated engineering, where the superb motivation behind distributed computing is to offer types of assistance.

III. TYPES OF TRUST MODEL

The convention, system, and techniques utilized for assessing the level of trust are known as trust models. The examination and recognizable proof of the trust model bases on their component. The trust model is partitioned into four sections

IV. SURVEY OF EXISTING TRUST MANAGEMENT MODEL



Title	Ref	Technique	+ve	Parameters
Trust Management Middleware for cloud service preference by prioritization assessment performed using covariance analysis	[19]	Objective trust evaluated using prioritized aggregation operation. Subjective trust	Weight is allotted to QoS in powerful way dependent on need class and fulfillment level.	Availability, reliability, security, privacy and customer support,

© Copyright: All rights reserved.

A Reliability based Trust management Mechanism for cloud service	[20]		Proposed mechanism is very simple and based on average method.	Reliability that based on consistency, familiarity
SMI Cloud: A framework for computing and ranking cloud service	[22]	Filter Mechanism is proposed based on historical feedback with frequency of usage with time instance. Proposed a CSMIC (Cloud Service measurement Index Consortium) model for judging a best resource provider dependent on fundamental and superfluous QoS according to consumer need. The proposed model utilized AHP strategy for appointing QoS.	Consumer specify their own essential and non-essential requirement according to their need with each QoS parameters have different dimension units(Boolean, Numeric, Unordered, Range)	Response time, Sustainability, Cost, Elastically, suitability, Accuracy, Stability, Availability, Usability, transparency, interoperability, Reliability, Adaptability.
Two Way Ranking Based Service Mapping in Cloud Environment	[23]	Proposed a model known as TRCSM based on perspective of both Service provider and consumer using AHP.	For service provider ranking model considered SMI and QoS of consumers. For consumer ranking QoS of service providers and behaviour attributes are considered.	QoS of service Provider are accountability, agility, confirmation, financial ,performance, security and protection, ease of use. QoS of Consumers are Turnover, Duration, and Transaction.
Towards A Trust Evaluation Middleware for cloud service selection	[18]	Proposed a model known as TRUSS that is a middleware approach for assessing the trustworthiness of resource provider based on both subjective as well as objective approach.	Reputation of consumer also evaluated for judging either consumer is trustworthy or not. QoS based on user preference	Objective trust dependent on QoS checking and SLA of administration and performance of service provider is monitored at time interval
A cloud service trust evaluation model based on combining weights and Gray correlation analysis	[24]	Proposed a cloud service trust evaluation model (CSTEM). Objective trust is evaluated by using rough set theory and subjective trust based on AHP(analytical hierarchical process)	Here penalty mechanism was introduced to control the malicious entities.	Cost, performance, reputation
Multi-level Trust Agreement in Cloud Environment Proposed	[25]	MLT A (multilevel trust agreement) framework that worked on hierarchical level and used the ABC((artificial Bee Colony) algorithm for evaluating trust.	The proposed model performs functions feedback assessment, risk observing, data accessibility, reward/ punishment selection, and time factor investigation.	Availability, time factor, risks monitoring, reward and punishment.

V. CONCLUSION

"In this paper, we have presented an outline of the cloud environment with its feature, deployment model, service model, basis entities of the cloud environment and the concerns related to security and privacy issues of consumers. The paper covered the present issue of cloud computing and why trust plays a crucial role in distributed computing. We discussed trust management techniques with their comparison, trust management system, types of trust, nature, and factors that affect the trust value. This paper presented a different trust assessment models with their QoS parameters for the distributed computing. The overview introduced in this article shows the fact that energizing advancement has been made toward a careful survey about the execution practices of different Cloud services, there are as yet a wide range of open issues in every classification that need further examination; consequently offering fascinating points for future exploration.

REFERENCES

1. T. H. Noor, Q. Z. Sheng, S. Zeadally, and J. Yu, "Trust Management of Services in Cloud Environments :," vol. 46, no. 1, pp. 1–30, 2013.
2. S. Pearson, Privacy , Security and Trust in Cloud Computing.
3. T. H. Noor, Q. Z. Sheng, S. Zeadally, and J. Yu, "Trust management of services in cloud environments," ACM Comput. Surv., vol. 46, no. 1, pp. 1–30, 2013, doi: 10.1145/2522968.2522980.

© Copyright: All rights reserved.

4. S. Pearson, Privacy , Security and Trust in Cloud Computing. .
5. X. Yang, S. Wang, B. Yang, C. Ma, and L. Kang, "A service satisfaction-based trust evaluation model for cloud manufacturing," Int. J. Comput. Integr. Manuf., vol. 32, no. 6, pp. 533–545, 2019, doi: 10.1080/0951192X.2019.1575982.
6. E. Kristiani, C. T. Yang, Y. T. Wang, and C. Y. Huang, "Implementation of an edge computing architecture using openstack and kubernetes," Lect. Notes Electr. Eng., vol. 514, pp. 675–685, 2019, doi: 10.1007/978-981-13-1056-0_66.
7. G. Aghaee, G. Mehran, and M. Ramin, "A new multi - level trust management framework (MLTM) for solving the invalidity and sparse problems of user feedback ratings in cloud environments," J. Supercomput., no. 0123456789, 2020, doi: 10.1007/s11227-020-03348-1.
8. P. Varalakshmi, T. Judgi, and D. Balaji, "Trust Management Model Based on Malicious Filtered Feedback in Cloud," Commun. Comput. Inf. Sci., vol. 804, pp. 178–187, 2018, doi: 10.1007/978-981-10-8603-8_15.
9. H. Kurdi, A. Alfaries, A. A. S. Alkharji, M. Addegaither, L. Altoaimy, and S. Hassan, "A lightweight trust management algorithm based on subjective logic for interconnected cloud computing environments," J. Supercomput., 2018, doi: 10.1007/s11227-018-2669-y.
10. S. based T. M. for C. E. Machhi, "Feedback based Trust Management for Cloud Environment," 2016.
11. S. S. Manvi and G. Krishna Shyam, "Resource management for Infrastructure as a Service (IaaS) in cloud computing: A survey," J. Netw. Comput. Appl., vol. 41, no. 1, pp. 424–440, 2014, doi: 10.1016/j.jnca.2013.10.004.
12. M. Chiregi and N. J. Navimipour, "A new method for trust and reputation evaluation in the cloud environments using the recommendations of opinion leader's entities and removing the effect of troll entities," Comput. Human Behav., vol. 60, pp. 280–292, Jul. 2016, doi: 10.1016/j.chb.2016.02.029.
13. R. Nagarajan, R. Thirunavukarasu, and S. Shanmugam, "A Fuzzy-Based Intelligent Cloud Broker with MapReduce Framework to Evaluate the Trust Level of Cloud Services Using Customer Feedback," Int. J. Fuzzy Syst., vol. 20, no. 1, pp. 339–347, 2018, doi: 10.1007/s40815-017-0347-5.
14. V. K. Damera, A. Nagesh, and M. Nagaratna, "Trust evaluation models for cloud computing," Int. J. Sci. Technol. Res., vol. 9, no. 2, pp. 1964–1971, 2020.
15. Q. Duan, "Cloud service performance evaluation: status, challenges, and opportunities – a survey from the system modeling perspective," Digit. Commun. Networks, vol. 3, no. 2, pp. 101–111, 2017, doi: 10.1016/j.dcan.2016.12.002.
16. S. Jaswal and M. Malhotra, "A detailed analysis of trust models in cloud environment," ACM Int. Conf. Proceeding Ser., pp. 1–5, 2019, doi: 10.1145/3368691.3368740.
17. M. Tang, X. Dai, J. Liu, and J. Chen, "Towards a trust evaluation middleware for cloud service selection," Futur. Gener. Comput. Syst., vol. 74, pp. 302–312, 2017, doi: 10.1016/j.future.2016.01.009.
18. M. B. Smithamol and S. Rajeswari, "TMM: Trust Management Middleware for Cloud Service Selection by Prioritization," J. Netw. Syst. Manag., vol. 27, no. 1, pp. 66–92, 2019, doi: 10.1007/s10922-018-9457-0.
19. W. Fan and H. Perros, "A Reliability-based Trust Management Mechanism for Cloud Services," 2013, doi: 10.1109/TrustCom.2013.194.
20. A. M. Al-faifi, B. Song, and M. Mehedi, "Performance prediction model for cloud service selection from smart data Performance prediction model for cloud service selection from smart data," Futur. Gener. Comput. Syst., no. March, 2018, doi: 10.1016/j.future.2018.03.015.
21. S. K. Garg, S. Versteeg, and R. Buyya, "SMICloud : A Framework for Comparing and Ranking Cloud Services," no. Vm, 2011, doi: 10.1109/UCC.2011.36.
22. N. Yadav and M. S. Goraya, "Two-way Ranking Based Service Mapping in Cloud Environment," Futur. Gener. Comput. Syst., 2017, doi: 10.1016/j.future.2017.11.027.
23. Y. Wang, J. Wen, X. Wang, B. Tao, and W. Zhou, "A cloud service trust evaluation model based on combining weights and gray correlation analysis," Secur. Commun. Networks, vol. 2019, 2019, doi: 10.1155/2019/2437062.
24. S. Kaushik and C. Gandhi, "Multi-level Trust Agreement in Cloud Environment," 2019 12th Int. Conf. Contemp. Comput. IC3 2019, pp. 1–5, 2019, doi:10.1109/IC3.2019.8844933.
25. https://en.wikipedia.org/wiki/Cloud_computing

Use one space after periods and colons. Hyphenate complex modifiers: "zero-field-cooled magnetization." Avoid dangling participles, such as, "Using (1), the potential was calculated." [It is not clear who or what used (1).] Write instead, "The potential was calculated by using (1)," or "Using (1), we calculated the potential."

Use a zero before decimal points: "0.25," not ".25." Use "cm^3," not "cc." Indicate sample dimensions as "0.1 cm × 0.2 cm," not "0.1 × 0.2 cm^2." The abbreviation for "seconds" is "s," not "sec." Do not mix complete spellings and abbreviations of units: use "Wb/m^2" or "webers per square meter," not "webers/m^2." When expressing a range of values, write "7 to 9" or "7-9," not "7~9."

A parenthetical statement at the end of a sentence is punctuated outside of the closing parenthesis (like this).

(A parenthetical sentence is punctuated within the parentheses.) In American English, periods and commas are within quotation marks, like "this period."

AUTHORS PROFILE

Ms. Pooja Goyal, MCA, M.Tech,UGC(NET) COMP.SCI, Pursuing Ph. D from MDU University Rohtak. Designation: Assistant Professor Department: Information TechnologyE-mail: poojagoyal895@gmail.com Area of research:- Cloud Computing.

Dr. Sukhvinder Singh Deora, MCA, M.Sc. (Mathematics), MPhil, PhD, NET. Designation: Asst. Professor, Dept. of Computer Science & Application, M.D. University, Rohtak

© Copyright: All rights reserved.